BLOEMFONTEIN CAMPUS

CSIS3764

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

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SEMESTER TEST 2
20 October 2023

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TIME: 180 min **MARKS:** 90

INSTRUCTIONS:

- This question paper has three questions:
  - Answer Question 1 on the answer sheet provided. (Closed Book)
  - Create and submit a Jupyter Notebook for Question 2. (Open Book)
  - Create and submit a Jupyter Notebook for Question 3. (Open Book)
- Submit the two notebooks to the following Blackboard links, respectively:
  - CSIS3764 -> Assessments -> Semester Test 2 -> Question 2.
  - CSIS3764 -> Assessments -> Semester Test 2 -> Question 3.

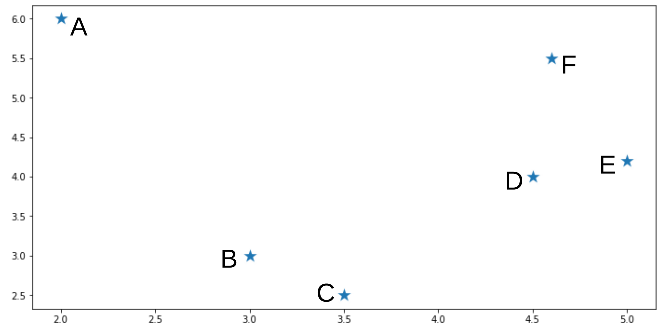
<u>Question 1 [20 Marks - 40 Minutes]</u> (Closed Book)

1.1 Consider the following learning curves of two machine learning models. On the left, a learning curve of a Support Vector Machine classifier and on the right, a learning curve of a K-Nearest Neighbour classifier:

1.1.1 Which model would most probably provide better prediction results? (1)

1.1.2 Motivate your answer in 1.1.1 by explaining why the relevant model would provide better prediction results. (3)

1.2 Consider the following scatter plot:



1.2.1 Draw a dendrogram to illustrate how the data points can be grouped by means of Agglomerative Hierarchical clustering. (5)

1.3 Explain the steps that are involved when the k-Means clustering algorithm is used to group data points into clusters. (5)

1.4 Briefly discuss the most important pre-processing steps when preparing text data for sentiment analysis. (6)

[20]

<u>Question 2 [50 Marks - 100 Minutes]</u> (Open Book)

Please log into Blackboard and go to CSIS3764 -> Assessments -> Semester Test 2 -> Question 2 and download the data file named *bank_customers.csv*. The data contains information about bank customers who own credit cards. Create a Jupyter notebook that contains the Python code to construct machine learning classifiers that are able to determine whether a current customer will stay with the bank or intends to leave, depending on the feature values for each specific customer. Name the notebook "*CSIS3764_YourStudentNumber_ST2-2_2023*".

The Jupyter notebook should have the following functionality (NB!!! - Show the changes in the data after each operation):

2.1 Import the data file into a dataframe named "*dfcus*" with the column headings as contained in the data file. (1)

2.2 Inspect the data by: (2)
- Generating a concise summary of the dataframe such as the index, data type, columns, non-null values and memory usage.
- Displaying the number of unique values for each feature.

2.3 Discard the first column, as well as last 7 columns of the dataset without explicitly specifying the column names. (2)

2.4 Generate a statistical summary of all the features. (1)

2.4.1 What can be deduced from the statistical summary with regard to the customers' marital status? (Print your answer in the Jupyter notebook) (2)

2.5 Visualise the data by using the seaborn library to create the following plots. Below each plot, discuss the information that can be deduced from the relevant plot: (Print your answer in the Jupyter notebook) (6)
- Create a categorical plot that shows the number of customers belonging to a specific Card Category for each Education Level.

- o Create a single combined scatter plot between the customers' Income Category and their Credit Limit for all the different Card Categories.

2.6 Determine whether the data is balanced with regard to the number of existing customers and attrited customers by: (4)

- o Creating a pie chart that shows the percentage of records for each customer group.
- o Percentages displayed on the pie chart should be rounded to 1 decimal.

2.7 Perform the necessary operations to balance the data: (5)

- o Use an undersampling technique to balance the number of records with regard to existing customers and attrited customers.
- o Assign the resampled data to the dataframe named "*dfcus_resampled*".

2.8 Convert the text values in the dataframe to numeric values: (7)

- o Convert the columns that contain only two unique text values to binary values [0, 1].
- o Convert the columns that contain more than two unique text values to a vector space using one-hot encoding.

2.9 Define X and y and create a training / testing dataset of 80% / 20%: (7)

- o X = Data features.
- o y = Attrition_Flag.
- o Determine the dimensions of X_train and X_test.
- o Ensure that all pre-processing is complete.

2.10 Train the following classifiers and determine the best model by using k-fold cross-validation: (8)

- o Naive Bayes (default values).
- o Support Vector Machine (gamma = auto).
- o Set k = 5 for the k-fold cross-validation.
- o Report the cross-validated training accuracy and F1 scores for all the classifiers.
- o Draw a learning curve for each of the classifiers.

2.11 Select the model that produced the highest F1 score to do the following: (3)

- o Make predictions using the test dataset.
- o Provide the test accuracy score, confusion matrix and classification report of the model.

2.12 Discuss the selected model's training accuracy in comparison to its testing accuracy. (Print your answer in the Jupyter notebook) (2)

[50]

<u>Question 3 [20 Marks - 40 Minutes]</u> (Open Book)

Please log into Blackboard and go to CSIS3764 -> Assessments -> Semester Test 2 -> Question 3 and download the data file named *wine.csv*. The data contains information about different types of wine with regard to their content.

Create a Jupyter notebook that contains the Python code to construct an unsupervised machine learning model that will be able to split different types of wine into different groups according to their content information. Name the notebook "*CSIS3764_YourStudentNumber_ST2-3_2023*".

The Jupyter notebook should have the following functionality (NB!!! - Show the changes in the data after each operation.):

3.1 Read the data file *wine.csv* into a dataframe named "*wine*". (1)

3.2 Make sure there is no *zero* values in the dataset. (1)

3.3 Use k-Means clustering to train a machine learning model that will have the capability to distinguish between different wines. (11)

- Use only the following features:
  - Alcohol
  - Magnesium
  - Total_Phenols
  - Flavanoids
  - Color_Intensity
- Make sure all pre-processing is complete.
- Determine the best value for k by calculating and plotting the silhouette score.
- Train the model, predict the labels and print the labels of the trained model.
- Print the cluster centres that were identified by the trained model.

3.4 Visualise the data using the seaborn package: (5)

- Create scatter plots between the different pairs of features. This should give you 10 separate scatter plots.
- Show the clusters that were identified by the model within the data by means of different colours.
- Also plot the centroids of the clusters on the same scatter plot.

3.5 Evaluate and discuss the optimal k-value that was obtained from the silhouette score in relation to the number of clusters that are visible within the scatter plots. (Print your answer in the Jupyter notebook) (2)

[20]

End of Paper